

Shantanu Shastri

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EDUCATION

Carnegie Mellon University

Master of Information Systems Management,

December 2026

GPA: 3.85

Manipal Institute of Technology

Bachelor of Electronics and Communications Engineering with Minor in Data Science,

August 2022

GPA: 3.99

Relevant Coursework: Introduction to Deep Learning, Generative AI Lab, LLM Applications, MLOps, Distributed Systems

SKILLS

Languages & Libraries: Python, Java, C++, JavaScript, TypeScript, SQL, NumPy, Pandas, Bash/Shell, PyTorch

AI/ML: TensorFlow, Hugging Face, LangChain, Prompt Engineering, RAG, Scikit-learn, MLflow

Infra & Database: AWS, GCP, Azure, Docker, FAISS, Terraform, Kafka, MySQL, MongoDB, Redis, Azure Blob, Tornado, RabbitMQ

Web: FastAPI, Flask, React.js, Redux, Node.js, GraphQL, REST API, gRPC, HTML/CSS

EXPERIENCE

BlackRock

Gurgaon, India

Software Developer 2

January 2025 – July 2025

Trade routings remodeling - Post Trade

- **Generative AI & LLM Engineering:** Engineered a pluggable micro-frontend interface using **Azure LLM APIs** to translate natural language into Java Query Language (JQL), **fine-tuning models** on a 40,000+ record corpus for high-precision code generation.
- **RAG & Model Optimization:** Implemented a **Retrieval-Augmented Generation (RAG)** pipeline using vector stores to enhance prompt context, boosting user acceptance rates from **68% to 82%** and **reducing query formulation time by 91%**.

Software Developer 1

July 2022 – December 2024

ML-based Recommender System

- **Data Pipeline Engineering:** Targeted routing inefficiencies across **60+ global markets** by analyzing and cleaning complex transactional data with **Python** and **Pandas**, uncovering latent patterns that optimized downstream routing efficiency.
- **Predictive Modeling:** Addressed high manual configuration costs by building a scalable **XGBoost** recommender workflow, achieving **high-80% accuracy** and reducing manual intervention by **60%**.

Multi-Instance – Multi-Threading

- **Distributed Systems Architecture:** Resolved critical singleton bottlenecks by **engineering a sharded, multi-instance server** environment with **Zookeeper leader election**, effectively distributing load and **boosting system performance by 24%**.
- **DevOps & Quality Engineering:** Built comprehensive **CI/CD pipelines** using **Azure DevOps**, with integrated **JUnit/Jacoco** testing suites, **reducing time-to-market by 40%** while maintaining **SONAR scores >90%**.

Shunya

June 2020 – September 2020

Data Science Intern

- **Computer Vision Automation:** Targeted inefficiencies in manual grading by engineering a GPU-accelerated **OpenCV** workflow in **Python** to analyze answer sheets, successfully automating the evaluation process for rapid feedback.
- **Adaptive Learning Algorithms:** Addressed learning gaps using **K-means clustering** to recommend targeted practice questions.

PROJECTS

Zulip Open-Source LLM Extensions

January 2026

- **LLM Feature Engineering:** Extended the Zulip platform by building a "Message Recap" engine using **Tornado, Redis, and RabbitMQ** that leverages **Large Language Models (LLMs)** to synthesize unread streams into concise, clickable summaries.
- **Cost & System Optimization:** Designed a cost-efficient **Topic Drift Detection system** to correct conversation titles in real-time, optimizing interaction models to significantly minimize token usage costs.

Real-Time Generative AI Product Agent

October 2025

- **Generative AI & Vector Search:** Developed an **AI agent** utilizing **Meta's FAISS** for high-speed vector retrieval to **generate high-fidelity product images** from text descriptions.

High-Performance Modular Booking Architecture

December 2025

- **Backend Scalability:** Engineered a decoupled **microservices architecture** using **Apache Kafka**, successfully handling bursts of **100,000+ real-time requests** with **minimal latency spikes**.
- **Infrastructure & DevOps:** Orchestrated **containerized deployments** using **Docker** to ensure environment consistency and enable **seamless horizontal scaling** across production systems.